1 Tipo espectral de las estrellas

1.1 El origen del espectro estelar

El siguiente parrafo , describe con gran detalle a que se debe el conjunto de lineas emitidas vistas en los espectros de las estrellas. Fuente: https://www.handprint.com/ASTRO/specclass.html

- Spectral classification examines only the light emitted by the relatively thin external layer or *photosphere* of a star.

- The thermodynamically stable stellar plasma below the **photosphere** creates a **continuous blackbody spectrum** that peaks at a specific temperature. This determines the overall shape of a star's flux profile — from the high energy, leftward rising profile of "blue" Type O and B stars that peak in the ultraviolet, through the relatively flat profile of "white" Type F and G stars that peak in the visual, to the low energy, rightward rising profile of "red" Type K and M stars that peak in the far red or infrared (diagram, right).

- The spectral type corresponds to the *effective temperature* ($\mathbf{T}_{\mathrm{e}}$) of a star, which is the blackbody temperature that matches its radiant flux per surface area — equivalent to the calculated temperature of the blackbody profile that best fits the overall shape and peak emittance of the observed flux profile.

- The photosphere decreases in temperature from the inside out: the outer surface of the photosphere is cooled by radiating energy into space, which allows it to absorb some of the light from the stellar interior.

- As the **temperature** of the exterior layer of the **photosphere goes down**, it is able to **absorb more energy** and display **more absorption features**. Increased absorption significantly "filters" or obscures the flux profile at low temperatures, especially in high luminosity stars and stars with metallic content. In these situations the ratio between the strength of specific absorption lines can be used to infer temperature.

- This method is used to calculate the *excitation temperature* ($\mathbf{T}_{\mathrm{x}}$), measured as the ratio in the strength of absorption lines produced by atoms of the same element at two different electron energies, or the *ionization temperature* ($\mathbf{T}_{\mathrm{i}}$), measured as the ratio in the strength of absorption lines produced by atoms of the same element stripped of one or more electrons.

- Because the peak excitation or ionization of different elements occurs at different temperatures, comparison of the line strength of different elements can be used to infer the photosphere effective temperature (spectral type).

- Es decir que el tipo espectral, esta dado por la temperatura efectiva de la fotosfera.

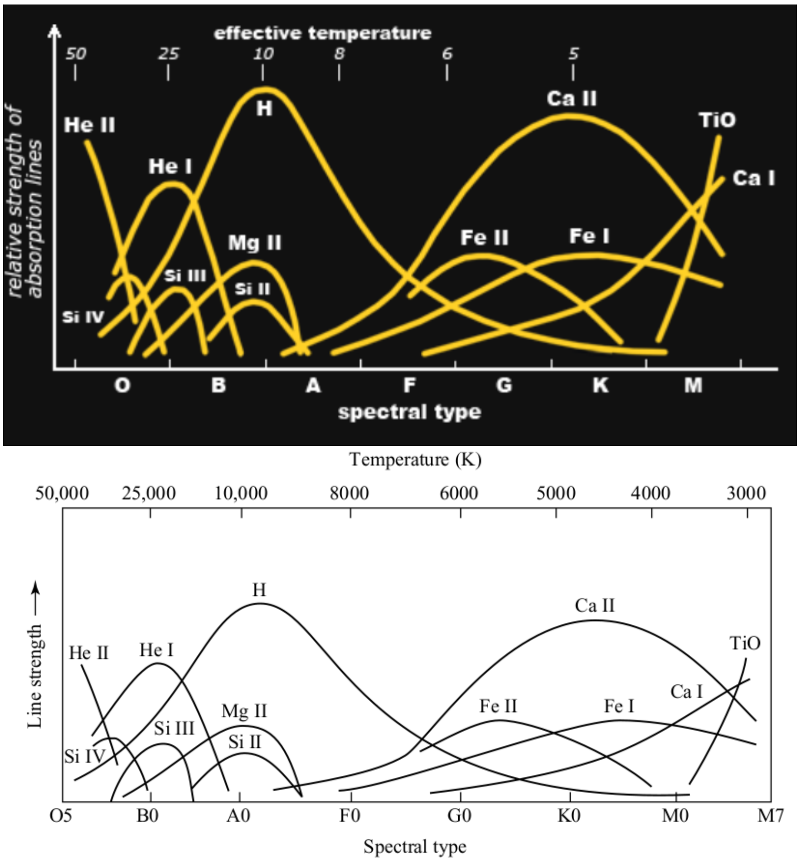


FIGURE 8.11 The dependence of spectral line strengths on temperature.

Figure 1. En esta figura podemos ver la intensidad de distintas lineas de absorcion, segun su tipo espectral y su temperatura efectiva. En la figura de abajo tenemos el diagrama de Carroll-Ostie donde se aprecia mejor la temperatura. A muy altas temperaturas podemos ver lineas de absorcion muy fuertes del He ionizado (He II). Para estrellas del tipo G en adelante puede verse una fuerte presencia del calcio Ionizado.

1.1.1 Descripcion general de lineas espectrales

- Hydrogen (H) lines (including the closely spaced absorption features that disappear at the Balmer Jump) are strongest in Type A0 stars (hence the original place of the **A** stars at the head of the stellar sequence), and grow fainter at both higher and lower temperatures.

- Sodium (Na) and calcium (Ca) lines become significant in Type G stars and grow more prominent as stars become cooler. Diatomic molecules including titanium oxide (TiO), cyanogen (CN) and many organic molecules form when the photosphere temperatures fall below a few thousand kelvins.

- In the coolest brown dwarf objects, water vapor (H_2O) and methane (CH_4) can form in the atmosphere.

1.2 Sistema de Clasificacion de Secchi

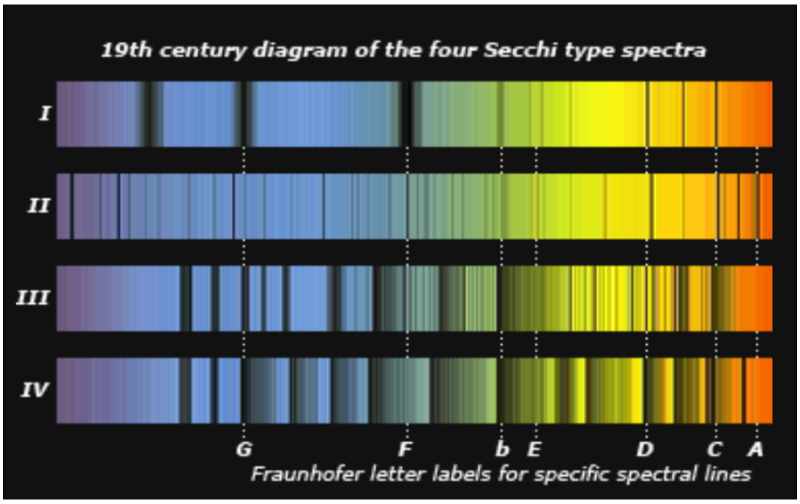


Figure 2. En lineas generales este esquema de clasificacion se basa en: Grupo I: Estrellas blancas, con lineas de H solamente. Grupo II: Estrellas amarillas, con espectros similares al solar. Grupo III: Estrellas rojas, con espectros similares al sol tambien + bandas moleculares en rojo [?]. Grupo IV: Estrellas muy rojas. Intensas bandas moleculares en la region azul [?].

1.3 Sistema de Clasificacion de Harvard

- El sistema de clasificacion de Harvard es una secuencia de ordenamiento de tipos estelares, de temperaturas decreciente que va desde las estrellas mas Calientes O [$\sim$30.000 [K]] hasta las mas frias, las M ($\sim$2.400[K]).

- Se agregan a este esquema de clasificacion las estrellas las L y T con temperaturas menores as 2.300[K]. Estas corresponden a las enanas marrones.

- Las estrellas, estan mayormente compuesta de Hidrogeno: *Thereafter Cecilia Payne-*

Gaposchkin (1925) identified the elemental ionization states produced by different surface temperatures and demonstrated that stars consist mostly of hydrogen

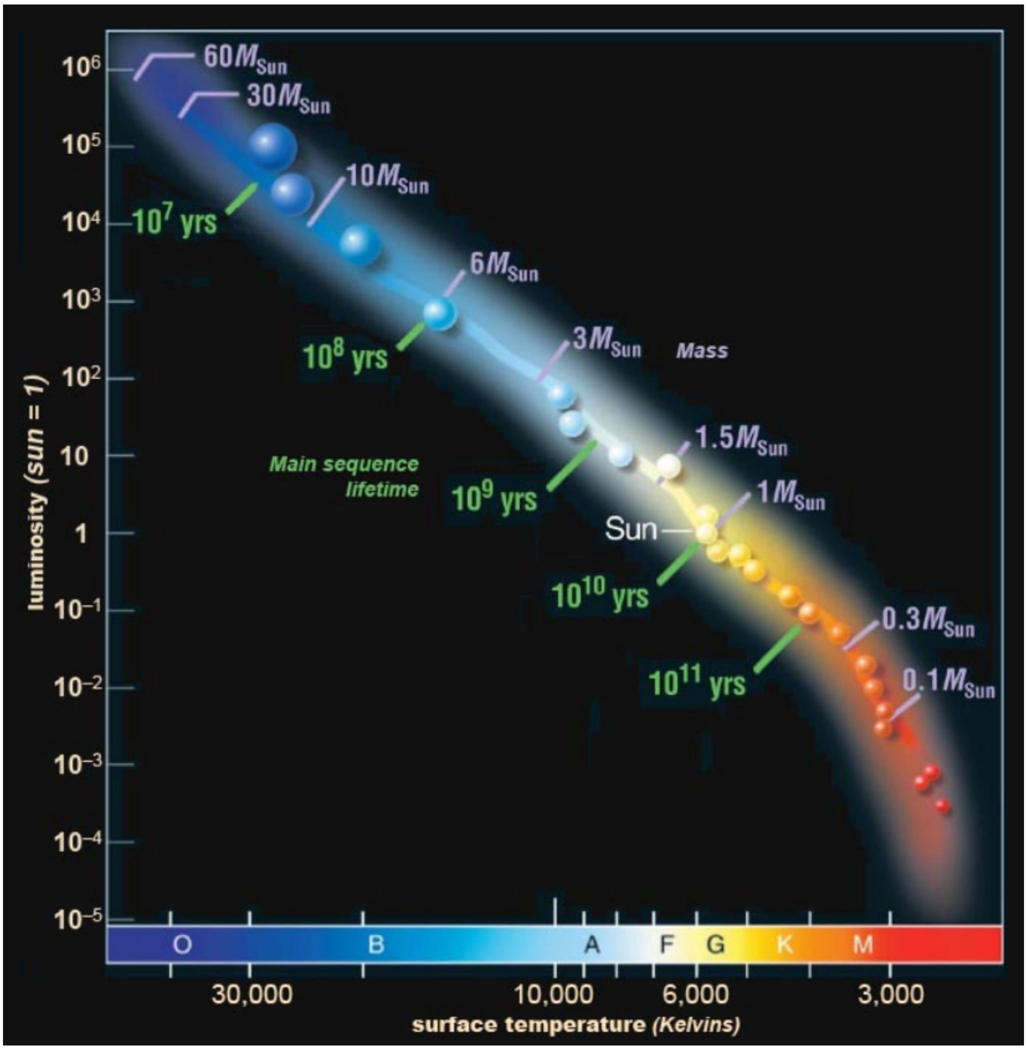


Figure 3. Figura que muestra el sistema de clasificacion de Harvard. En el eje horizontal puede verse la clasificacion del mismo y las regiones que abarca cada uno, con sus colores.

Class	Effective temperature	Conventional color description	Actual apparent color
O	≥ 30,000 K	blue	blue
B	10,000–30,000 K	blue white	deep blue white
A	7,500–10,000 K	white	blue white
F	6,000–7,500 K	yellow white	white
G	5,200–6,000 K	yellow	yellowish white
K	3,700–5,200 K	orange	pale yellow orange
M	2,400–3,700 K	red	light orange red
L	1,300–2,400 K	red brown	scarlet
T	500–1,300 K	brown	magenta
Y	≤ 500 K	dark brown	black

Figure 4. Otro esquema con las temperaturas y colores del sistema de clasificacion de Harvard.

TABLE 8.1 Harvard Spectral Classification.

Spectral Type	Characteristics
O	Hottest blue-white stars with few lines Strong He II absorption (sometimes emission) lines. He I absorption lines becoming stronger.
B	Hot blue-white He I absorption lines strongest at B2. H I (Balmer) absorption lines becoming stronger.
A	White Balmer absorption lines strongest at A0, becoming weaker later. Ca II absorption lines becoming stronger.
F	Yellow-white Ca II lines continue to strengthen as Balmer lines continue to weaken. Neutral metal absorption lines (Fe I, Cr I).
G	Yellow Solar-type spectra. Ca II lines continue becoming stronger. Fe I, other neutral metal lines becoming stronger.
K	Cool orange Ca II H and K lines strongest at K0, becoming weaker later. Spectra dominated by metal absorption lines.
M	Cool red Spectra dominated by molecular absorption bands, especially titanium oxide (TiO) and vanadium oxide (VO). Neutral metal absorption lines remain strong.
L	Very cool, dark red Stronger in infrared than visible. Strong molecular absorption bands of metal hydrides (CrH, FeH), water (H_2O), carbon monoxide (CO), and alkali metals (Na, K, Rb, Cs). TiO and VO are weakening.
T	Coolest, Infrared Strong methane (CH_4) bands but weakening CO bands.

S and C spectral types for evolved giant stars are discussed on page 466.

Figure 5. Las estrellas O tienen lineas de Absorcion de He, tanto de He neutro (He I) y ionizado (He II). En las B, tenemos He neutro y H neutro. A partir de las siguientes clases aparece el calcio ionizado Ca II.

2 El diagrama de HERTZSPRUNG–RUSSELL

- En un principio, se intento establecer una conexion entre las magnitudes/luminosidades y los tipos espectrales. Lo que se observo es que habia distintos valores de luminosidades y magnitudes absolutas para estrellas de un mismo tipo espectral.
- Las diferencias en la luminosidad son causadas por la edad de la estrella (o estadio evolutivo). Hay que tener en cuenta que se habia desarrollado una teoria donde las estrellas iban perdiendo temperatura a medida que se acercaban al final de su estadio evolutivo. Esta idea ya no es una idea acpetada - Carroll-Ostie:
  - These regularities led to a theory of stellar evolution that described how stars might cool off as they age.
  - This theory (no longer accepted) held that stars begin their lives as young, hot, bright blue O stars.
  - It was suggested that as they age, stars become less massive as they exhaust more and more of their “fuel” and that they then gradually become cooler and fainter until they fade away as old, dim red M stars.
  - Although incorrect, a vestige of this idea remains in the terms early and late spectral types.

2.1 Que sucede en la vida de una estrella?

Fuente: https://www.handprint.com/ASTRO/specclass.html

- After a main sequence star has burned (through nuclear fusion) enough hydrogen to accumulate a massive core of helium "ash" (which takes about 10 billion years for a solar mass star), this core collapses under its own weight, increasing the core temperature to the point where helium fusion can occur and the resulting thermal pressure counteracts the gravitational contraction.

- The star fuses helium into carbon inside a shell of still burning hydrogen. In the most massive stars, a similar collapse of carbon ash fuses to form oxygen, which combines to form heavier elements — neon, sodium, magnesium, sulfur, silicon and **finally metals up to iron**, which **cannot release energy through nuclear fusion**. (Energy is not released through iron fusion but must be added to make it happen, which means *chemical elements heavier than iron are only formed in the energetic collapse and explosion of a* ***supernova***, where the necessary energy is supplied by a catastrophic gravitational collapse.)

- In the devolution of an aging star, the outward pressure of energy released by the fusion of heavier elements is able to balance the inward force of gravity.

- But at some point the mass of the core ash is insufficient to compress the core into the temperature necessary to fuse heavier elements, and gravity wins.

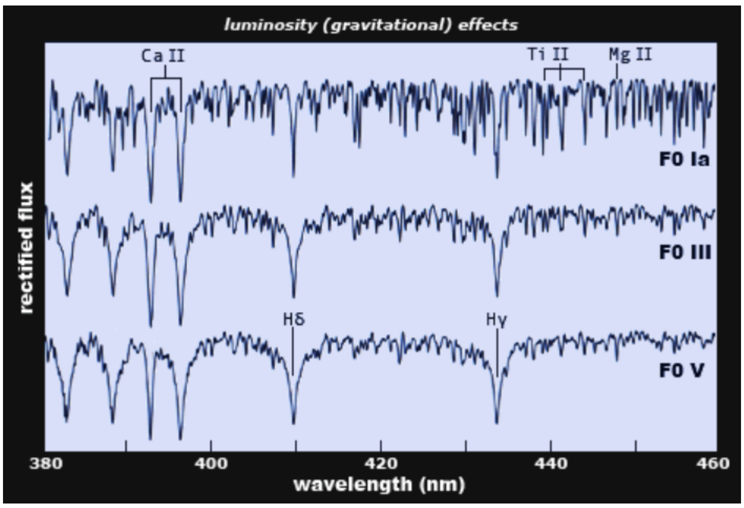
- In solar mass stars, this final collapse produces a nova that results in an inert and slowly cooling white dwarf inside a planetary nebula; in the most massive stars the gravitational collapse of the iron core produces a supernova that results in a neutron star or black hole.

- Before that end stage collapse occurs, the increased **energy outflow** produced by the transition to helium fusion **causes** a dramatic **increase in the radius** of the star.

- Because the **photosphere has expanded** to a great distance from the nuclear core and also has a greatly increased surface area, its **temperature goes down**.

- In addition, because it is much farther from the center of mass, the gravitational pressure on the photosphere is reduced: it expands and becomes more rarefied (density decreases).

- This cooler, larger and less dense photosphere causes a **narrowing** and **strengthening of spectral lines** and the **appearance** of many **absorption lines** from any "metals" that were part of the star's original composition (diagram , bottom).



- The **increased surface area** of the star also **increases the luminosity** of the photosphere and the **absolute magnitude** of the star.

- These larger, brighter stars are *subgiant, giant* or *supergiant* stars, which can be tens to hundreds of times the radius and emit hundreds to thousands of times the energy of main sequence stars of the same mass. These spectral changes are indicated as the Morgan–Keenan luminosity class - joined to the spectral type.

2.2 Morgan–Keenan Luminosity Classes

- The **Morgan–Keenan (MK) luminosity classes** are part of a stellar classification system that categorizes stars not just by their **spectral type** (which tells us temperature and color) but also by their **luminosity**, which is related to their **size and stage in stellar evolution**. This system helps astronomers understand where a star is in its lifecycle.

- The MK system was developed in the 1940s by **William Wilson Morgan** and **Philip C. Keenan**. The full MK classification combines:

  - A **spectral type** (O, B, A, F, G, K, M — from hottest to coolest)

  - A **luminosity class** (Roman numerals I through V, and beyond)

- Together, they describe both a star's **surface temperature** and its **absolute brightness** (luminosity), and thus indirectly its **radius**.

- Luminosity classes reflect differences in **stellar surface gravity**, which in turn is related to **stellar radius**:

  - **Supergiants (I)** have **low surface gravity**, meaning they're very large and expanded.

  - **Dwarfs (V)** like the Sun have **high surface gravity**, since they're compact.

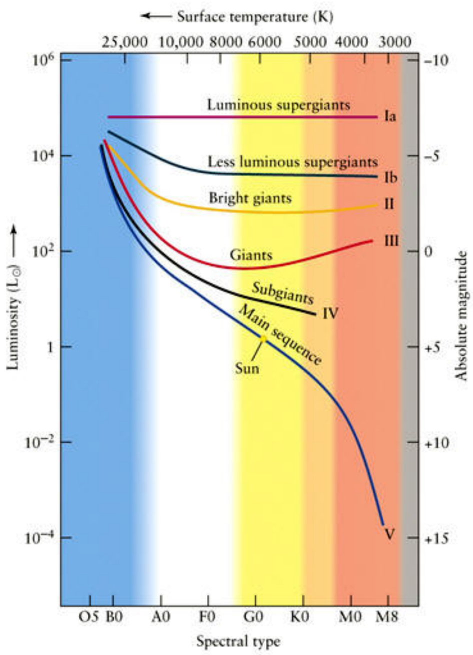


Figure 6.

🌟 **Luminosity Classes Overview**

Class	Type	Description	Examples
0	Hypergiants	Extremely luminous, rare	VY Canis Majoris
Ia	Bright Supergiants	Most luminous supergiants	Rigel (B8 Ia)
Iab	Intermediate Supergiants	Between Ia and Ib	Betelgeuse (M2 Iab)
Ib	Less luminous supergiants		
II	Bright Giants	Not quite supergiants	
III	Giants	Evolved stars, post-main-sequence	Arcturus (K1.5 III)
IV	Subgiants	Between main sequence and giants	
V	Main Sequence (Dwarfs)	Normal hydrogen-burning stars	Sun (G2 V), Vega (A0 V)
VI	Subdwarfs	Hotter, less luminous than dwarfs	
VII	White Dwarfs	Stellar remnants	Sirius B (DA2)

TABLE 8.3 Morgan–Keenan Luminosity Classes.

Class	Type of Star
Ia-O	Extreme, luminous supergiants
Ia	Luminous supergiants
Ib	Less luminous supergiants
II	Bright giants
III	Normal giants
IV	Subgiants
V	Main-sequence (dwarf) stars
VI, sd	Subdwarfs
D	White dwarfs

Figure 7.

3 Ecuaciones Importantes

- Relacion (Aproximada) Masa - Luminosidad:

  $\frac{L}{L_\odot} = \left(\frac{M}{M_\odot}\right)^{3.5}$

- Relacion luminosidad-temperatura-radio:

  $\log\left(\frac{L}{L_\odot}\right) = 2\log\left(\frac{R}{R_\odot}\right) - 4\log\left(\frac{T_e}{T_{e,\odot}}\right)$

4 Notas

4.1 Sobre el desarrollo en cuanto a la teoria de evolucion estelar y estrellas en general

- By that time fundamental advances had been made in astrophysics. These included:

- Max Planck's equations for blackbody radiation using a quantum theory of energy (1901),
- Niels Bohr's electron shell theory of atomic structure (1913),
- Elucidation of the mass/luminosity relationship by Ejnar Hertzsprung and H.N. Russell (1905-1913),
- Description of atomic ionization states by Megh Nad Saha (1920).
- Cecilia Payne-Gaposchkin (1925) identified the elemental ionization states produced by different surface temperatures and demonstrated that stars consist mostly of hydrogen;
- Sir Arthur Eddington applied Karl Schwarzchild's concept of radiative equilibrium to the *The Internal Constitution of the Stars* (1916-1925);
- Hans Bethe (1939) described the nuclear fusion processes that generate stellar energy, which Subrahmanyan Chandrasekhar (1939) formalized as a theory of stellar evolution, including the gravitational collapse that produces supernovae and white dwarfs.